Polar Curves and Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Converting between polar and rectangular coordinates**

Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.

Multiply by r: $r^2 = 4r\cos\theta \Rightarrow x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0 \Rightarrow (x-2)^2 + y^2 = 4$. This is a circle of radius 2 centered at $(2, 0)$.

2 **[Foundational]** • **Target: Identifying and sketching common polar curves**

Identify the type of curve for $r = 2 + 2\sin\theta$, and state its key feature.

This is a cardioid (since the coefficient of $\sin\theta$ equals the constant term) — a heart-shaped curve passing through the pole.

3 **[Foundational]** • **Target: Identifying and sketching common polar curves**

Identify the type of curve for $r = \cos(3\theta)$.

This is a rose curve with 3 petals (odd n in $r = \cos(n\theta)$ gives n petals).

4 **[Foundational]** • **Target: The slope of a polar curve**

State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r\cos\theta$ and $y = r\sin\theta$.

$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$.

5 **[Foundational]** • **Target: Finding horizontal and vertical tangents**

For the circle $r = 4\cos\theta$, find the value(s) of θ in $[0, \pi]$ where the tangent line is horizontal (where $y = r\sin\theta$ has a critical point).

$y = 4\cos\theta\sin\theta = 2\sin(2\theta)$. $\frac{dy}{d\theta} = 4\cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$.

6 **[Foundational]** • **Target: Symmetry of polar curves**

Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the x-axis).

Replace θ with $-\theta$ in the equation. If the equation is unchanged (or yields the same curve), the curve is symmetric about the polar axis. For example, $r = 2 + 2\cos\theta$ satisfies $f(-\theta) = 2 + 2\cos(-\theta) = 2 + 2\cos\theta = f(\theta)$, confirming symmetry.

7 **[Foundational]** • **Target: Symmetry of polar curves**

Test whether $r = 3\sin\theta$ is symmetric about the y-axis (replace θ with $\pi - \theta$).

$f(\pi - \theta) = 3\sin(\pi - \theta) = 3\sin\theta = f(\theta)$ (using $\sin(\pi - \theta) = \sin\theta$). Symmetric about the y-axis.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: More polar-to-rectangular conversions**

Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0$, $0 \le \theta < 2\pi$).

$r = \sqrt{3^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = \sqrt{36} = 6$. $\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{3}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$ (in Quadrant I, matching both coordinates positive).

9 **[Intermediate]** • **Target: More polar-to-rectangular conversions**

Convert the rectangular equation $y = x$ to polar form.

$r\sin\theta = r\cos\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ (a constant-θ ray, i.e. a straight line through the pole).

10 **[Intermediate]** • **Target: More slope-of-a-polar-curve practice**

Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin\theta$ at $\theta = 0$.

$\frac{dr}{d\theta} = \cos\theta$. $\frac{dy}{dx} = \frac{\cos\theta\sin\theta + r\cos\theta}{\cos\theta\cos\theta - r\sin\theta}$. At $\theta = 0$: $r = 1$, $\frac{dr}{d\theta} = 1$. Numerator: $1(0) + 1(1) = 1$. Denominator: $1(1) - 1(0) = 1$. $\frac{dy}{dx} = \frac{1}{1} = 1$.

11 **[Intermediate]** • **Target: More slope-of-a-polar-curve practice**

Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle θ, and interpret the result using the fact that a circle centered at the origin has radial symmetry.

$\frac{dr}{d\theta} = 0$. $\frac{dy}{dx} = \frac{0 \cdot \sin\theta + 5\cos\theta}{0 \cdot \cos\theta - 5\sin\theta} = \frac{5\cos\theta}{-5\sin\theta} = -\cot\theta$. This matches the slope of the tangent line to a circle at angle θ — perpendicular to the radius at that point, consistent with basic circle geometry.

12 **[Intermediate]** • **Target: Tangent lines at the pole**

Find the values of θ where the cardioid $r = 1 - \cos\theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.

$1 - \cos\theta = 0 \Rightarrow \cos\theta = 1 \Rightarrow \theta = 0$. At the single angle where $r = 0$, the curve touches the pole; the tangent line there is along $\theta = 0$ (the polar axis), since that's the direction the curve approaches the origin from.

13 **[Intermediate]** • **Target: Tangent lines at the pole**

Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.

$\cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ — these are the four tangent-line directions at the pole for this 4-petaled rose.

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14 [Intermediate] • Target: Classifying polar curves by equation type

Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):

a $r = 5$ Circle centered at the pole

b $r = 3 + 3\cos\theta$ Cardioid (coefficient equals constant)

c $r = 1 + 3\sin\theta$ Limaçon with an inner loop (constant $<$ coefficient)

d $r = 4\sin(5\theta)$ Rose with 5 petals (odd $n = 5$)

(a) Circle centered at the pole; (b) Cardioid (coefficient equals constant); (c) Limaçon with an inner loop (constant $<$ coefficient); (d) Rose with 5 petals (odd $n = 5$)

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 6.02 Taylor and Maclaurin Series

State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether n is odd or even.

If n is odd, the rose has exactly n petals (traced over $[0, \pi]$). If n is even, the rose has $2n$ petals (requires the full $[0, 2\pi]$ to trace, since each petal is retraced with opposite sign after π).

16 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 1.01 Integration by Parts

How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?

$r = 2\sin(4\theta)$: $n = 4$ (even) $\Rightarrow 2(4) = 8$ petals. $r = 5\cos(7\theta)$: $n = 7$ (odd) $\Rightarrow 7$ petals.

17 [Challenging] • Target: Graphing polar curves point-by-point • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Complete a short table of (θ, r) values for $r = 2 + 2\cos\theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.

$\theta = 0$: $r = 4$. $\theta = \frac{\pi}{2}$: $r = 2$. $\theta = \pi$: $r = 0$. $\theta = \frac{3\pi}{2}$: $r = 2$. The curve is widest ($r = 4$) along the polar axis, pinches to a point ($r = 0$) at $\theta = \pi$ (opposite direction), forming the characteristic heart shape.

18 [Challenging] • Target: Concavity considerations for polar curves • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Explain, in terms of $\frac{dy}{dx}$, what a 'vertical tangent' means for a polar curve, and why it corresponds to $\frac{dx}{d\theta} = 0$ (with $\frac{dy}{d\theta} \neq 0$).

A vertical tangent means the slope $\frac{dy}{dx}$ is undefined (infinite) — since $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$, this happens exactly when the denominator $\frac{dx}{d\theta} = 0$ while the numerator stays nonzero, making the ratio blow up.

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19 [Challenging] • Target: More slope-of-a-polar-curve practice • Cross-Lesson: 3.01 Parametric Equations

Level 2 · MediumFind the angle(s) $\theta \in [0, \pi]$ at which the polar curve $r = 3\cos\theta$ passes through the origin, and determine the equation of the tangent line at that point.

Set $r = 0$: $3\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$. By the Tangents at the Pole Theorem, when $r(\theta_0) = 0$ and $r'(\theta_0) eq 0$, the line $\theta = \theta_0$ is the tangent line. Thus, the tangent line at the pole is $\boldsymbol{\theta = \frac{\pi}{2}}$ (the vertical line $x = 0$).

20 [Challenging] • Target: Tangent lines at the pole • Cross-Lesson: 3.02 Vector-Valued Functions

Level 2 · MediumFind the slope of the tangent line $\frac{dy}{dx}$ to the polar cardioid $r = 1 - \cos\theta$ at $\theta = \frac{\pi}{2}$.

Recall $x = r\cos\theta$ and $y = r\sin\theta$. By the Product Rule: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$. Given $r = 1 - \cos\theta \Rightarrow r' = \sin\theta$. At $\theta = \pi/2$: $r = 1 - 0 = 1$, $r' = 1$, $\sin(\pi/2) = 1$, $\cos(\pi/2) = 0$. Substitute: $\frac{dy}{dx} = \frac{(1)(1) + (1)(0)}{(1)(0) - (1)(1)} = \frac{1}{-1} = \boldsymbol{-1}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting between polar and rectangular coordinates

How many petals does the rose $r = \cos(2\theta)$ have?

a. 4 ✓ **b.** 6

c. 7 **d.** 2

Here $n = 2$, which is even, so the rose has $2n$ petals: 4 petals.

22 [Foundational] • Target: Identifying and sketching common polar curves

Find the smallest positive θ where the rose $r = 3\sin(4\theta)$ passes through the pole.

a. $\frac{\pi}{4}$ ✓ **b.** $\frac{\pi}{16}$

c. $\frac{3\pi}{4}$ **d.** $\frac{\pi}{8}$

Set $3\sin(4\theta) = 0 \Rightarrow \sin(4\theta) = 0$. The smallest positive solution is $4\theta = \pi$, so $\theta = \frac{\pi}{4}$.

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23 [Foundational] • Target: The slope of a polar curve

The limaçon $r = 7 + 14\cos\theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{2\pi}{3}$ ✓

b. $\frac{\pi}{2}$

c. $\frac{5\pi}{6}$

d. $\frac{4\pi}{3}$

Set $r = 0$: $7 + 14\cos\theta = 0 \Rightarrow \cos\theta = -\frac{1}{2}$. The smaller positive solution is $\theta = \frac{2\pi}{3}$.

24 [Foundational] • Target: Finding horizontal and vertical tangents

What is the slope of the line described by the polar equation $\theta = \pi/6$?

a. $-\frac{\sqrt{3}}{3}$

b. 0

c. $\sqrt{3}$

d. $\frac{\sqrt{3}}{3}$ ✓

$\theta = \pi/6$ is the ray/line through the origin with slope $\tan(\pi/6) = \frac{\sqrt{3}}{3}$.

25 [Foundational] • Target: Symmetry of polar curves

Find r for the rectangular point $(20,\ 21)$.

a. 41

b. 841

c. **29** ✓

d. 1

$r = \sqrt{x^2 + y^2} = \sqrt{20^2 + 21^2} = 29$.

26 [Foundational] • Target: More polar-to-rectangular conversions

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?

a. **4** ✓

b. 2

c. 3

d. 6

The slope of the tangent line to the graph of $f(x)$ at any point is given by its derivative, $f'(x)$. 1. Find the derivative of $f(x) = x^2 - 2x - 1$ using the power rule: $f'(x) = 2x - 2$ 2. Evaluate the derivative at the x -coordinate of point $P(3, 2)$, which is $x = 3$: $f'(3) = 2(3) - 2 = 6 - 2 = 4$ Thus, the slope of the tangent line at P is 4. Misconception Analysis: - 2 results from mistakenly reading the y-coordinate of point P as the slope, since $f(3) = 2$. - 3 results from taking the x-c.

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27 [Foundational] • Target: More slope-of-a-polar-curve practice

Convert $r = 28\cos\theta$ to rectangular form and identify the radius of the resulting circle.

a. 56

b. 196

c. 28

d. **14** ✓

Multiply by r: $r^2 = 28r\cos\theta \Rightarrow x^2 + y^2 = 28x \Rightarrow (x-14)^2 + y^2 = 196$. Radius $= 14$.

28 [Foundational] • Target: Tangent lines at the pole

Find the area of one petal of the rose $r = 3\sin(2\theta)$ (integrate θ from 0 to $\frac{\pi}{2}$).

a. $\frac{9\pi}{4}$

b. $\frac{9\pi}{16}$

c. $\frac{9\pi}{8}$ ✓

d. 9π

Area $= \frac{1}{2}\int_0^{\frac{\pi}{2}} [3\sin(2\theta)]^2\, d\theta = \frac{9\pi}{8}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Classifying polar curves by equation type

Rose curve $r = \cos(n\theta)$: number of petals when n is even:

a. n

b. $2n$ ✓

c. $n + 1$

d. $4n$

Applying Parametric and Polar principles, Even n: $2n$ petals. Therefore, the verified correct option is $2n$.

30 [Intermediate] • Target: Number of petals in rose curves

Rose curve $r = \cos(n\theta)$: number of petals when n is odd:

a. n ✓

b. $2n$

c. $n/2$

d. $4n$

Applying Parametric and Polar principles, Odd n: n petals. Therefore, the verified correct option is n.

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31 [Intermediate] • Target: Graphing polar curves point-by-point

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.

a. 17

b. **21** ✓

c. 9

d. 11

The area of the shaded region is given by the definite integral of $f'(x)$ from $x = 1$ to $x = 5$. This area is also equal to the change in the function $f(x)$ over this interval: $\int_1^5 f'(x)\,dx = f(5) - f(1)$. The graph shows that the function $f'(x)$ is a constant function, represented by the horizontal line $y = 4$. Therefore, $f'(x) = 4$. The area of the shaded region is given as 16 square units. Using the definite integral: $\int_1^5 4\,dx = [4x]_1^5 = 4(5) - 4(1) = 20 - 4 = 16$. This con.

32 [Intermediate] • Target: Concavity considerations for polar curves

Find the area of one petal of $r = 1\cos(3\theta)$ (integrate θ from $-\frac{\pi}{6}$ to $\frac{\pi}{6}$).

a. $\frac{\pi}{12}$ ✓

b. π

c. $\frac{\pi}{6}$

d. $\frac{\pi}{24}$

Area $= \frac{1}{2}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} [1\cos(3\theta)]^2\,d\theta = \frac{\pi}{12}$.

33 [Intermediate] • Target: Converting between polar and rectangular coordinates

A sector of the circle $r = 10$ sweeps from $\theta = 0$ to $\theta = \pi/6$. Find its area.

a. $\frac{50\pi}{3}$

b. $\frac{25\pi}{3}$ ✓

c. $\frac{25\pi}{6}$

d. $\frac{5\pi}{3}$

Area $= \frac{1}{2}\int_0^{\pi/6} 10^2\,d\theta = \frac{1}{2} \cdot 100 \cdot \pi/6 = \frac{25\pi}{3}$.

34 [Intermediate] • Target: Identifying and sketching common polar curves

Find the area of the sector of $r = 3$ from $\theta = 0$ to $\theta = \pi/4$.

a. $\frac{31\pi}{24}$

b. $\frac{9\pi}{4}$

c. $\frac{9\pi}{8}$ ✓

d. $\frac{9\pi}{16}$

Area $= \frac{1}{2}\int_0^{\pi/4} 3^2\,d\theta = \frac{1}{2} \cdot 9 \cdot \pi/4 = \frac{9\pi}{8}$.

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35 [Intermediate] • Target: The slope of a polar curve

Find the total area enclosed by the cardioid $r = 2(1 + \cos\theta)$.

a. 6π ✓

b. 3π

c. 4π

d. 12π

Area $= \frac{1}{2}\int_0^{2\pi} [2(1+\cos\theta)]^2\, d\theta = \frac{3}{2}\pi \cdot 4 = 6\pi$.

36 [Intermediate] • Target: Finding horizontal and vertical tangents

Find the area inside $r = 11$ but outside $r = 6$.

a. 121π

b. 85π ✓

c. 25π

d. $\frac{85\pi}{2}$

Area $= \pi 11^2 - \pi 6^2 = \pi(11^2 - 6^2) = 85\pi$.

37 [Intermediate] • Target: Symmetry of polar curves

The slope field for a differential equation is shown above. The differential equation is of the form $\frac{dy}{dx} = ky$ for some constant k. If the solution curve passes through the point $(0, 1)$ and $(1, 2)$, find the value of k.

a. $\ln 2$ ✓

b. 2

c. $\frac{1}{2}$

d. 1

The differential equation is given by $\frac{dy}{dx} = ky$. This is a separable differential equation. We can rewrite it as $\frac{dy}{y} = k\,dx$. Integrating both sides, we get $\int \frac{dy}{y} = \int k\,dx$, which yields $\ln|y| = kx + C_1$. Exponentiating both sides, we have $|y| = e^{kx + C_1} = e^{C_1}e^{kx}$. Let $C = \pm e^{C_1}$, so $y = Ce^{kx}$. We are given that the solution curve passes through the point $(0, 1)$. Substituting these values into the general solution: $1 = Ce^{k(0)}$ $1 = Ce^0$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: More polar-to-rectangular conversions • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Parametric tangent at $t = t_0$: slope $m = \frac{y'(t_0)}{x'(t_0)}$. If $x'(t_0) = y'(t_0) = 0$, the point is:

a. always a cusp

b. regular

c. a node

d. **a singular/cusp point — need L'Hôpital** ✓

Applying Parametric and Polar principles, Both zero: singular point. Therefore, the verified correct option is a singular/cusp point — need L'Hôpital.

39 [Challenging] • Target: More slope-of-a-polar-curve practice • Cross-Lesson: 3.01 Parametric Equations

Tangent line to parametric curve is horizontal when:

a. $dx/dt = 0$

b. $dy/dt = 0$ **and** $dx/dte0$ ✓

c. $t = 0$

d. $dy/dt = dx/dt$

Applying Parametric and Polar principles, Horizontal tangent: $dy/dt = 0$. Therefore, the verified correct option is $dy/dt = 0$ and $dx/dte0$.

40 [Challenging] • Target: Tangent lines at the pole • Cross-Lesson: 3.02 Vector-Valued Functions

Tangent line to parametric curve is vertical when:

a. $dy/dt = 0$

b. $dx/dt = 0$ **and** $dy/dte0$ ✓

c. $dx/dt = dy/dt$

d. $t = 0$

Applying Parametric and Polar principles, Vertical tangent: $dx/dt = 0$. Therefore, the verified correct option is $dx/dt = 0$ and $dy/dte0$.

41 [Challenging] • Target: Classifying polar curves by equation type • Cross-Lesson: 4.02 Area in Polar Coordinates

At which point does $f(x) = x^2 - 4x$ have a horizontal tangent?

a. $x = 0$

b. $x = 2$ ✓

c. $x = 4$

d. $x = -2$

Applying Derivatives principles, $f'(x) = 2x - 4 = 0 \Rightarrow x = 2$. Therefore, the verified correct option is $x = 2$.

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42 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Tangent lines at origin to $r = \sin 3\theta$ (3-petal rose):

a. $\theta = 0, \pi/3, 2\pi/3$ ✓

b. $\theta = 0, \pi/6, \pi/3$

c. $y = 0$ only

d. 3 lines at $\theta = 0, \pi/3, 2\pi/3$

Rose passes through origin at $\sin 3\theta = 0$: $\theta = 0, \pi/3, 2\pi/3$.

43 [Challenging] • Target: Graphing polar curves point-by-point • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For the differential equation $\frac{dy}{dx} = 1x - 5y$, find the slope of the solution curve at the point $(3, 2)$.

a. -9

b. -5

c. 7

d. **-7** ✓

Substitute $x = 3$, $y = 2$: $\frac{dy}{dx} = 1(3) - 5(2) = -7$.

44 [Challenging] • Target: Concavity considerations for polar curves • Cross-Lesson: 6.01 Power Series and Interval of Convergence

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.

a. **1** ✓

b. 2

c. 3

d. 4

The equation of the tangent line to $f(x)$ at $x = 1$ is given as $y = 2x + c$. We are given that $f(1) = 3$. This means the point $(1, 3)$ lies on both the function $f(x)$ and its tangent line at $x = 1$. Since the point $(1, 3)$ lies on the tangent line $y = 2x + c$, we can substitute the coordinates into the equation: $3 = 2(1) + c$ $3 = 2 + c$ $c = 3 - 2$ $c = 1$ The derivative of the function at $x = 1$ is the slope of the tangent line, which is 2. This information is consistent with the problem but not direct.

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45 [Challenging] • Target: Converting between polar and rectangular coordinates • Cross-Lesson: 6.02 Taylor and Maclaurin Series

The slope of $y = 3x - 5$ at any point is:

a. -5

b. 0

c. **3** ✓

d. undefined

Applying Derivatives principles, Slope of a line $y = mx + b$ is $m = 3$. Therefore, the verified correct option is 3.

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